Environmental fluoxetine promotes skin cell proliferation and wound healing.

by Roa Hanoun and Jero Eyras

Abstract Pharmaceutical pollutants, including the antidepressant fluoxetine (FLX, commercial name: Prozac), are increasingly detected in aquatic environments, raising concerns about their effects on non-target organisms, including humans. The original study demonstrated that environmentally-relevant FLX concentrations promote wound closure in human skin biopsies and keratinocytes through enhanced cell proliferation and serotonin signalling, supported by transcriptomic, phosphoproteomic, and kinase profiling analyses. Building on these findings, the present study extends the transcriptomic analysis by examining gene expression in scratched HaCaT keratinocytes exposed to 540 ng/L FLX at two time points: 6 and 36 hours. Differentially expressed genes (DEGs) were identified at each time point and compared to unexposed control cells, with an intersection analysis performed between both conditions. We identified 147 DEGs at 6 hours and 250 DEGs at 36 hours, with upregulated genes associated with cell proliferation and cellular stress defence mechanisms. The difference in DEG numbers compared to the original study reflects sample quality filtering applied during data processing. These results provide a more detailed temporal resolution of FLX-induced transcriptomic changes and support the conclusion that environmental FLX exposure modulates gene expression in ways that may meaningfully impact skin wound healing.

Introduction

Pharmaceutical pollution has become a growing environmental concern, particularly the presence of antidepressants in aquatic ecosystems. Fluoxetine (FLX), commercially known as Prozac, is one of the most widely prescribed selective serotonin reuptake inhibitors (SSRIs) and works by blocking the reabsorption of serotonin in the brain, thereby increasing serotonin availability and improving mood regulation. Due to incomplete metabolisation in the human body and inadequate removal by wastewater treatment plants, FLX is consistently detected in surface waters at concentrations ranging from nanograms to micrograms per litre R Core Team (2012)(Metcalfe et al., 2010).

Serotonin is not only a neurotransmitter but also plays a well-established role in peripheral tissues, including the skin, where it is involved in regulating cell proliferation and wound healing. Wound healing is a complex and tightly coordinated biological process that proceeds through three overlapping phases: inflammation, proliferation, and tissue remodelling. During the proliferation phase, keratinocytes — the dominant cell type of the outermost skin layer (the epidermis) — migrate toward the wound edge and actively divide to restore the skin barrier. Any disruption or enhancement of this process can have significant clinical implications. Given that FLX modulates serotonin signalling, and serotonin itself plays a role in keratinocyte activity, it is biologically plausible that environmental FLX exposure could meaningfully influence skin repair mechanisms.

The original study by (Rodriguez-Barucg et al., 2024) was the first to investigate this link experimentally, using a multi-omics approach combining transcriptomics, phosphoproteomics, and kinase profiling. Their transcriptomic analysis identified 350 differentially expressed genes (DEGs) in FLX-exposed keratinocytes, with upregulated genes linked to cell proliferation and tissue morphogenesis, and downregulated genes associated with mitochondrial function and metabolism (Rodriguez-Barucg et al., 2024). These findings provided a molecular basis for the observed enhancement in wound closure, but left room for deeper temporal investigation of how gene expression changes evolve over time during FLX exposure.

Transcriptomics is the large-scale study of all RNA molecules expressed in a cell or tissue at a given moment, essentially providing a snapshot of which genes are active and to what degree. By comparing gene expression between exposed and unexposed cells, it is possible to identify which biological processes are being switched on or off in response to a treatment. In this study, we use RNA sequencing (RNA-seq) data analysed with DESeq2, a statistical tool specifically designed to detect differentially expressed genes from count-based sequencing data while accounting for biological variability between samples.

The present study focuses exclusively on the transcriptomic component of the original research. We analysed gene expression in scratched HaCaT human keratinocytes exposed to 540 ng/L FLX at two time points: 6 and 36 hours, mirroring the time points used in the transcriptomic section of the original study (Rodriguez-Barucg et al., 2024). The 6-hour time point captures the early transcriptional response to FLX exposure, reflecting immediate cellular adjustments, while the 36-hour time point represents a later stage where more sustained gene expression changes associated with active wound closure are expected. Our central research question is: does FLX exposure significantly alter gene expression in human keratinocytes in ways that support enhanced wound healing, and how does this response evolve between 6 and 36 hours of exposure? To address this, DEGs were identified at each time point and compared to unexposed control cells, followed by enrichment analysis to map these genes to relevant biological pathways. An intersection analysis was further performed to identify genes consistently differentially expressed across both time points. We also explored whether enrichment analysis could reveal additional gene functions and pathways beyond those reported in the original article (Rodriguez-Barucg et al., 2024).

Materials and Methods

Most likely this will be the largest section in your report (depending on the number of images you show in the `Results` section). Take a look at your experiments article to see how they organised its content to see if this is also suitable for your report.

This section must at the very least contain information about the **data**; how many samples from what organism + strain (if applicable), what kind of data is there available (not only saying **FPKM** but also the sequencer type that generated this data). Also, information about all the **software** that you used during your analysis (i.e. `R` and its libraries, `bioConductor`, `DESeq2`, etc.) and all **methods** that you applied (both filtering steps and statistical analysis steps). Try to use the proper terms and symbols if needed, i.e. "*We filtered for DEGs with an* $\alpha < 0.05$". Symbols need to be surrounded with `$` signs, otherwise you will get an error when knitting.

Do not show any results in this section, for instance when stating that you filtered t-test p-values on a certain value, you do not list how many genes were left, that is for the next section.

Cell samples

HaCaT cells…

RNA Sequencing

Illumina and else

Data analysis

ggplot (data visualization)

Data description RNA sequence output, 2 files, x columns…

Data cleaning Poisson distance, heatmap…

DEGs

DeSeq2 (pval, padj), enrichment (pval)

Results

Data Analysis

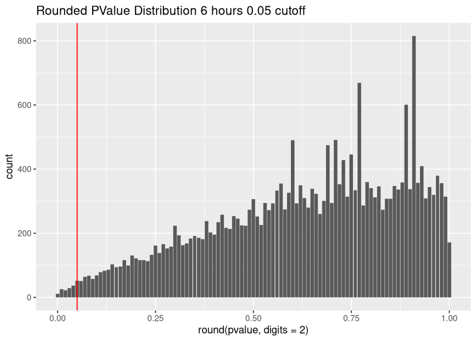
plots

After generating heatmaps and MDS plots, the most irregular samples identified are: C36-R1 and E36_R3 from the 36-hour dataset, and C6_R4 from the 6-hour dataset. These samples don't seem grouped and are quite far apart from the rest. This is the reason we chose to continue without those samples in the following DEGs section. From the 4 samples groups 3 have now only 3 replicates and the 4rth one remains with 4.

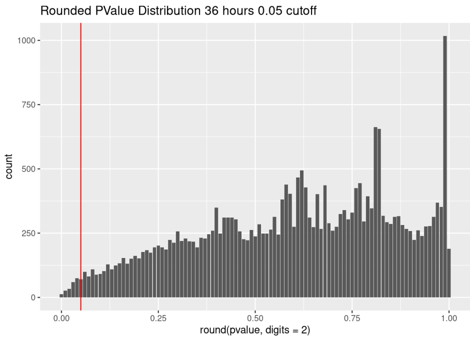
DEGs

No significant differentialy expressed genes where found using a p-adjusted value `padj < 0.05`. Thus, we proceed with p-values, although with the warning that we are not using p-adjusted values. Some significance can be seen in the following p-value distribution plots for both samples.

pval plots



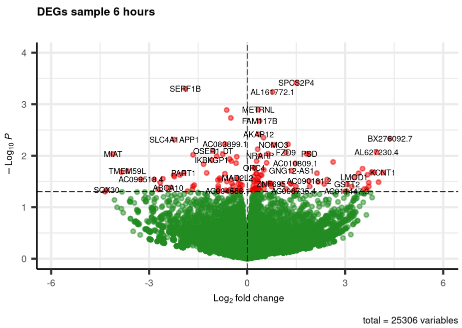
(a) A



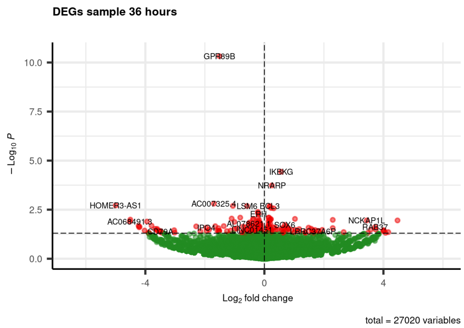
(b) B

With a p-value cutoff of 0.05 we end up with a similar number of DEGs as the original article. Of course, due to the data cleaning we end up with a lower amount, while without cleaning it would result in the exact number. The following volcano plots showcase the resulting DEGs.

volcano plots



(a) A



(b) B

Enrichment analysis

go plots

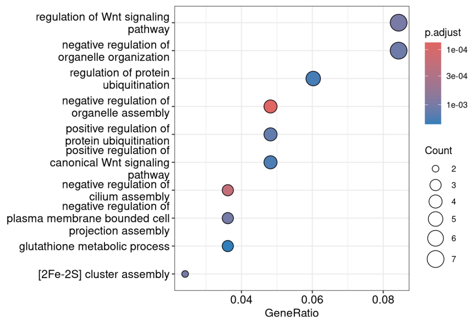
kegg plots

reactome plots

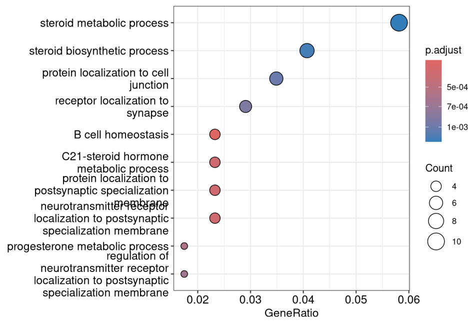
Pathview

Here you will only focus on your own results, start at the beginning but only show the interesting results. This section works best if you alternate between text introducing some result, a figure or table showing the result, and a short text segment analyzing the presented result *and* introducing the next figure, etc.

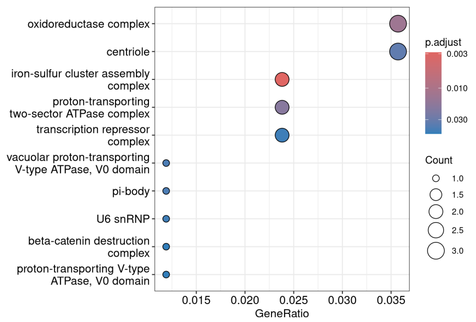
Although we have worked in `R` a lot, the results are almost never pieces of `R` code. If you have written your own algorithm or analysis method try to present it in a formula *followed* by the `R` implementation, otherwise this section should mainly contain figures and tables.



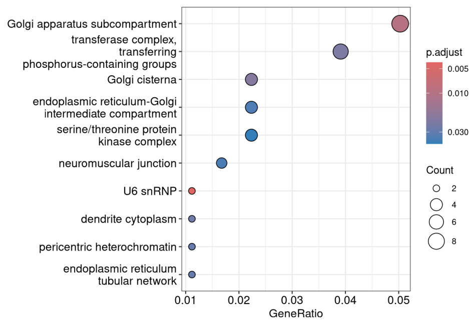
(a) A



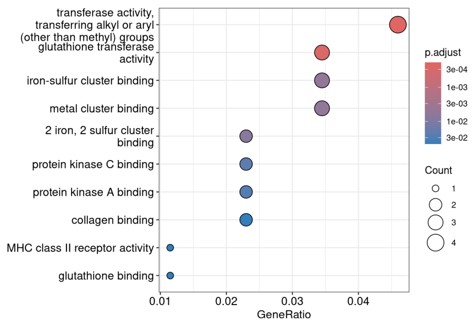
(b) B



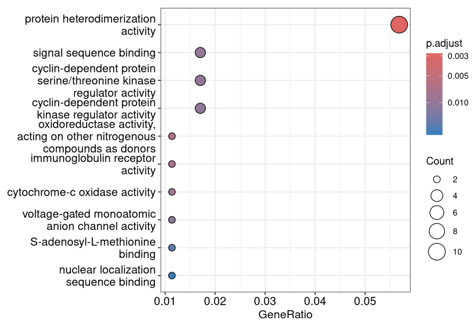
(c) A



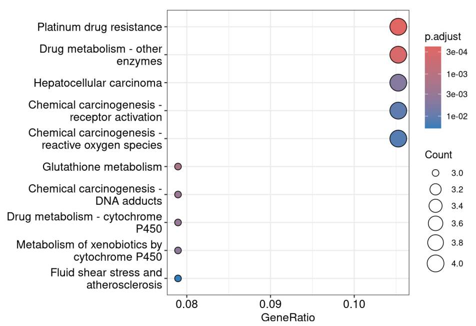
(d) B



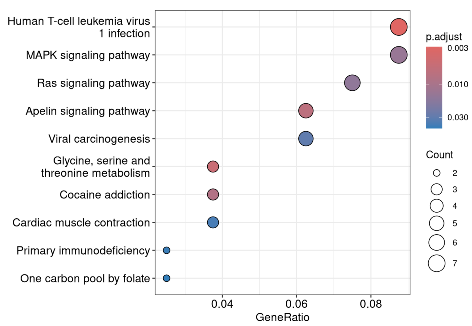
(e) A



(f) B

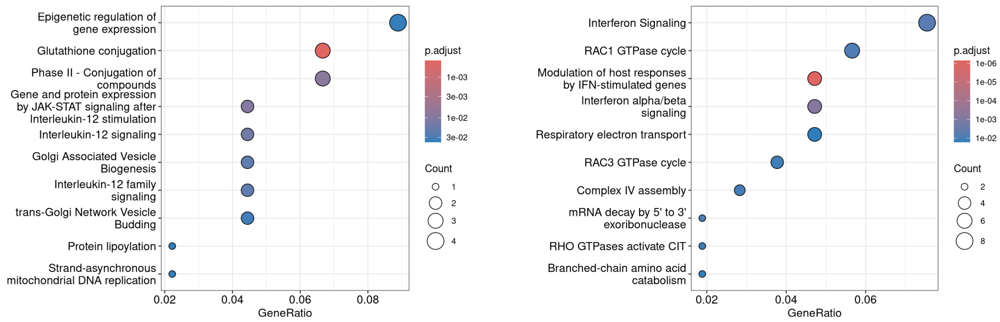


(a) A



(b) B

There are multiple methods of including a figure in this section. Figure 6 shows the LaTeXmethod (see the source code) giving very fine control over size and placement of a figure. The size of a figure is determined by the width of the text column, Figure 6 is scaled as 85% of the text width. The Wikibooks.org website shows some examples for placement (for instance, what the [htbp] means), scaling and captions.



(a) A **(b)** B

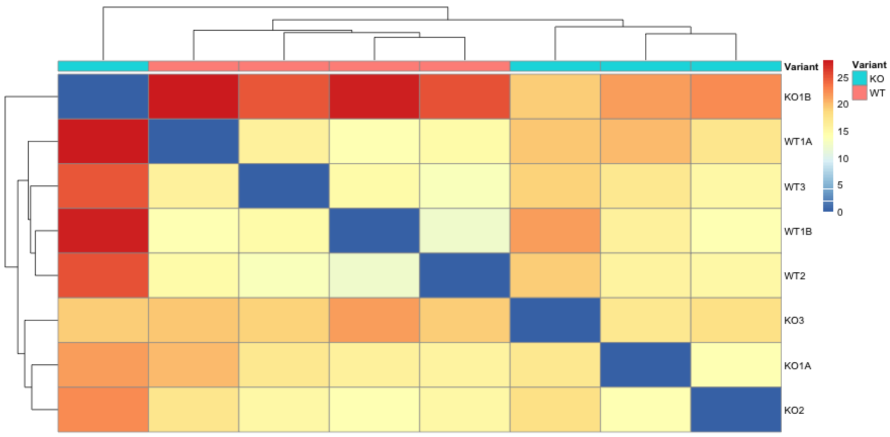
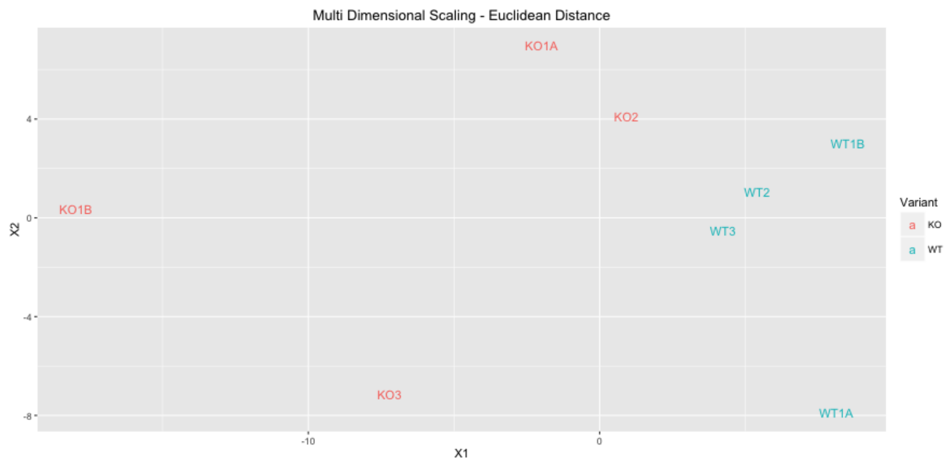


Figure 6: Heatmap showing sample distances and clustering.

Alternatively the previously shown R method using the PNG library can be used which allows changing the *heigth* and *width* of an image but lacks the possibility of referencing the figure in the text, aligning the figure (the heatmap is centered), providing a figure caption and automatic numbering of figures.

```
library(png)
library(grid)
img <- readPNG("images/mds.png")
grid.raster(img)
```



Conclusion

Limit to your *own* conclusions, based on what is shown in the **Results** section. Do not include any comparison to the original article as that is the content of the next section, the **Discussion**. Usually you refer back to all steps outlined in the results section since you included it there for a reason.

Discussion

An important section in your report; here you can compare your findings with the findings presented in the original article. If you found exactly the same results - and that was your goal from the start - this will be a bit shorter section and you should focus on other parts of your research that hasn't been discussed or shown in the article, i.e. any findings from the exploratory data analysis or differences in DEGs using the manual- and library-method.

Besides comparing the two 'experiments', this is also the place to discuss how reliable your results are. Did anything go wrong or did you get unexpected outcomes using some method and if so, what did you do to solve it or are there still open discussion points on this subject.

Notes on the `bibliography` section

The last section in the report contains all the used references that are used throughout the text in a section called the `Bibliography`. In the text you can link to references using the LaTeXcommands (see the source of this file for instructions). These references must be present in a file named `RJreferences.bib` which is automatically generated once you click the `Knit` button for the first time in a folder with the same name as the current markdown file (`./report-template/RJreferences.bib` for this example file).

This file containing the references can be edited in RStudio too, browse to the folder (after Knitting) and click on the `.bib` file to edit. If, for example, I want to add the article that is available for the experiment used in the week 4 - 7 assignment document, I would first lookup the paper in PubMed to get the details and add these details to the `.bib` file as follows:

```
@article{garcia,
    author = {Duart-Garcia, Braunschweig},
    title = {Functional expression study of igf2 antisense transcript in mouse},
    journal = {International Journal of Genomics},
    doi = {10.1155/2014/390296},
    url = {http://dx.doi.org/10.1155/2014/390296},
    year = 2014
}

@article{rodriguez,
    author = {Quentin Rodriguez-Barucg, Angel A. Garcia, Belen Garcia-Merino, Tomilayo Akinm
    title = {Environmental fluoxetine promotes skin cell proliferation and wound healing},
    journal = {Environmental Pollution},
    doi = {10.1016/2024/124952},
    url = {https://doi.org/10.1016/j.envpol.2024.124952},
    year = 2024
}
```

References added to this file do not automatically show up in your document, unless you actually refer to it. To add a reference, like so: (Duart-Garcia, 2014), you need to use the `\citep{}` or `\citer{}` LaTeXcommand combined with the given ID which is `garcia` in our case.

Bibliography

B. Duart-Garcia. Functional expression study of igf2 antisense transcript in mouse. *International Journal of Genomics*, 2014. doi: 10.1155/2014/390296. URL http://dx.doi.org/10.1155/2014/390296. [p6]

R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria, 2012. URL http://www.R-project.org/. ISBN 3-900051-07-0. [p1]

Q. Rodriguez-Barucg, A. A. Garcia, B. Garcia-Merino, T. Akinmola, T. Okotie-Eboh, T. Francis, E. Bringas, I. Ortiz, M. A. Wade, A. Dowle, D. A. Joyce, M. J. Hardman, H. N. Wilkinson, and P. Beltran-Alvarez. Environmental fluoxetine promotes skin cell proliferation and wound healing. *Environmental Pollution*, 2024. doi: 10.1016/2024/124952. URL https://doi.org/10.1016/j.envpol.2024.124952. [p1, 2]

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